Acyclicity of the Birhizomic Projection to the Face Category

Abstract

We prove that, for every monoid P and every presheaf of abelian groups F on B_P^{op}, the adjunction morphism

$$F \longrightarrow R\pi_*\pi^{-1}F$$

is a quasi-isomorphism. The proof is objectwise on B_P. The two inputs are: Verdier's hypercovering theorem for cohomology of a site, and the local normal form of birhizomic neighbourhoods, which says that the birhizomic hypercovers of an affine monoid have a cofinal subsystem indexed by good star covers of the associated conical face stratification. The carrier functor from such a star cover to the face category is initial; hence cofinality for derived inverse limits reduces the calculation to the face category, which has an initial object.

1 Conventions and notation

We use the convention natural for affine fan sites: a sheaf on A_P^{op} is written as a covariant functor on A_P. Thus an object of A_P is a homomorphism $P \to Q$, and a covering family of the object $P \to Q$ is a family of homomorphisms $Q \to Q_i$ satisfying the birhizomic valuative criterion.

For $x \in P$, write

$$b_x := P[-x]^\sharp.$$

Thus b_x is the initial sharp saturated monoid under P in which x becomes 0. Let $B_P \subset A_P$ be the full subcategory spanned by the objects $P \to b_x$.

If S is a monoid, we write

$$\mathsf{A}(S) := A_S^{\mathrm{op}}, \qquad \mathsf{B}(S) := B_S^{\mathrm{op}}.$$

Let

$$j_S : \mathsf{B}(S) \hookrightarrow \mathsf{A}(S)$$

be the inclusion. The associated morphism of topoi is denoted

$$\pi_S : \mathrm{Sh}(\mathsf{A}(S)) \longrightarrow \mathrm{Sh}(\mathsf{B}(S)).$$

When $S = P$ we write simply π.

2 Formal reductions

Lemma 2.1 (Slices)**.** *Let $b = b_x = P[-x]^\sharp \in B_P$. Then the slice site $(A_P^{\mathrm{op}})/b$ is canonically equivalent to A_b^{op}. Under this equivalence, B_P^{op}/b identifies with B_b^{op}. Consequently, for every sheaf G on A_P^{op},*

$$(R\pi_*G)(b) \simeq R\Gamma\big(A_b^{\mathrm{op}},\, G|_{A_b^{\mathrm{op}}}\big).$$

Proof. An object of $(A_P^{\mathrm{op}})/b$ is an object $P \to Q$ of A_P together with a morphism $Q \to b$ in A_P^{op}, equivalently a morphism $b \to Q$ in A_P. This is precisely an object of A_b. The same argument identifies B_P/b with B_b, because

$$b[-\bar{y}]^\sharp \simeq P[-x-y]^\sharp$$

for every image $\bar{y} \in b$. The formula for derived direct image is the usual description of the stalk over an object of a site as cohomology on the slice site; see SGA 4, Exp. V, Prop. 5.1. □

Thus it suffices to prove the following affine acyclicity statement.

Proposition 2.2 (Affine acyclicity)**.** *Let S be any monoid. For every presheaf of abelian groups $F\colon B_S \to \mathbf{Ab}$, viewed as a sheaf on B_S^{op} for the trivial topology, the natural map*

$$F(S) \longrightarrow R\Gamma\big(A_S^{\mathrm{op}},\, \pi_S^{-1}F\big)$$

is a quasi-isomorphism.

The proof occupies the next two sections.

3 Birhizomic hypercovers and their carrier functors

We recall the local form of the birhizomic topology. For a fine saturated sharp monoid S, let σ_S be the rational polyhedral cone

$$\sigma_S = \mathrm{Hom}(S, \mathbb{R}_{\geq 0}).$$

Its faces correspond to objects of B_S: a face $F \subset S$ gives the quotient S/F, equivalently some $S[-x]^\sharp$ after choosing x in the relative interior of F. If S is not finitely generated, all statements below are understood on finitely generated saturated submonoids; finite presentation of the maps involved makes this reduction harmless, and filtered colimits are exact in $\mathbf{Ab}$.

The following proposition is the monoid-theoretic input.

Proposition 3.1 (Local normal form for birhizomic neighbourhoods)**.** *Let S be a monoid. There is a cofinal system of hypercovers*

$$U_\bullet \longrightarrow S$$

of the final object of the site A_S^{op} with the following properties.

(i) To $U_\bullet$ is associated a small category I_U, the category of non-empty finite intersections of the members of the cover and their iterated faces.

(ii) There is a functor

$$\rho_U : I_U \longrightarrow B_S,$$

called the carrier functor. If $\alpha \in I_U$, then $\rho_U(\alpha)$ is the unique object $S[-x_\alpha]^\sharp$ determined by the face of S which is forced to vanish on the corresponding birhizomic neighbourhood.

(iii) For every presheaf $F\colon B_S \to \mathbf{Ab}$, the Cech–Verdier complex of $\pi_S^{-1}F$ on $U_\bullet$ is naturally the Roos complex computing

$$R\varprojlim_{I_U} F \circ \rho_U.$$

(iv) The carrier functor ρ_U is initial: for every $b \in B_S$, the comma category $(\rho_U \downarrow b)$ has contractible nerve.

Proof. We give the standard proof, since this is the only point at which the definition of "birhizomic" is used.

First assume that S is fine saturated. A homomorphism

$$v : S \longrightarrow V$$

to a valuative monoid determines a centre face

$$F(v) = \{s \in S \mid v(s) = 0\}.$$

After quotienting by $F(v)$, the non-zero values of $S/F(v)$ have a largest archimedean class; this gives a point in the relative interior of a face of $\sigma_{S/F(v)}$. Iterating gives the usual valuative flag of faces. The relation $x \prec y$ records precisely that x belongs to the group generated by the face containing y.

Let W be a rational open star in a sufficiently fine barycentric subdivision of the conical stratification of σ_S. Choose finitely many inequalities

$$a_i - b_i > 0$$

which cut out W in the relevant archimedean quotient, and finitely many elements c_j whose vanishing cuts out the carrier face. Define S_W to be the sharp saturation of the monoid obtained from S by

- forcing the c_j to become 0, and
- adjoining the differences $a_i - b_i$.

The map $S \to S_W$ is of finite presentation. We claim that it is birhizomic at exactly those valuative points whose visible archimedean point lies in W.

Indeed, suppose $S \to M \xrightarrow{p} N$ is a factorization with p infinitesimal, and suppose the valuative point lies in W. For an adjoined difference $a_i - b_i$, its image in N is positive and is not infinitesimal relative to the image of $a_i + b_i$. In M^{gp} one has

$$a_i - b_i \prec a_i + b_i,$$

and in N one has

$$p(a_i + b_i) \prec p(a_i - b_i).$$

By the defining property of an infinitesimal morphism, $a_i - b_i \in M$. Thus all adjoined generators lift. If c_j has value 0, then both c_j and $-c_j$ are forced in the same way; since M is sharp this forces $c_j = 0$. Hence the factorization through N lifts uniquely to S_W.

Conversely, if one of the required strict archimedean inequalities fails, one constructs an infinitesimal thickening by taking, inside the value group, the submonoid consisting of all elements whose leading visible value is positive, together with the image of S. This submonoid is saturated and sharp, the map to the given valuative monoid is infinitesimal, but the failed difference does not belong to it. Therefore no lift to S_W exists. Similarly, if a forced vanishing condition fails, the corresponding quotient cannot be lifted. This proves the claim.

Rational open stars form a basis for the birhizomic topology. Given any birhizomic cover, finite presentation shows that only finitely many strict archimedean inequalities and vanishing conditions occur locally. Refining the associated finite conical decomposition by a barycentric subdivision, we

obtain a star cover refining the given cover. Repeating this in each degree gives a cofinal system of hypercovers.

For such a star hypercover, intersections of stars are again unions of stars. The category I_U is the category of non-empty finite intersections. The value of the left Kan extension of F on the corresponding monoid is exactly $F(\rho_U(\alpha))$; sheafification does not change these values on the hypercover because the stars are birhizomic-acyclic by the preceding lifting criterion. Hence the Cech–Verdier complex is the Roos complex of $F \circ \rho_U$.

It remains to check initiality. Fix $b = S[-x]^\sharp \in B_S$, corresponding to a face τ of the cone. The comma category $(\rho_U \downarrow b)$ is the category of stars whose carriers specialize to τ. Geometrically this is the face category of a good star cover of the closed star of τ. That closed star is a convex polyhedral cone, hence contractible; by the nerve lemma, the nerve of $(\rho_U \downarrow b)$ is contractible. Thus ρ_U is initial.

For a general saturated sharp monoid S, every finite stage of a finite-presentation hypercover descends to a fine saturated submonoid $S_0 \subset S$. The preceding construction applied to S_0, followed by saturated base change to S, gives the required star hypercovers of S. Since the indexing system of such S_0's is filtered and filtered colimits are exact in abelian groups, the same cofinality and acyclicity statements hold for S. □

We shall also use the following standard cofinality theorem.

Lemma 3.2 (Cofinality for derived limits)**.** *Let $u : J \to I$ be an initial functor, meaning that $(u \downarrow i)$ has contractible nerve for every $i \in I$. Then, for every functor $E : I \to \mathbf{Ab}$, the natural map*

$$R\varprojlim_I E \longrightarrow R\varprojlim_J E \circ u$$

is an isomorphism in the derived category.

Proof. This is the derived form of the usual cofinality theorem for limits. It is equivalent to Quillen's Theorem A applied to the Grothendieck construction of E, or to SGA 4, Exp. I, Prop. 8.1.3. □

4 Proof of affine acyclicity

Proof of Proposition 2.2. Let $F : B_S \to \mathbf{Ab}$ be given. By Verdier's hypercovering theorem (SGA 4, Exp. Vbis, Th. 7.4.1), the cohomology

$$R\Gamma(A_S^{\mathrm{op}}, \pi_S^{-1} F)$$

is computed as the filtered colimit over any cofinal system of hypercovers of the corresponding Cech–Verdier complexes. We use the cofinal system of Proposition 3.1.

For a hypercover $U_\bullet$ in that system, Proposition 3.1 identifies the Cech–Verdier complex with

$$R\varprojlim_{I_U} F \circ \rho_U .$$

Since $\rho_U : I_U \to B_S$ is initial, Lemma 3.2 gives

$$R\varprojlim_{I_U} F \circ \rho_U \simeq R\varprojlim_{B_S} F .$$

But B_S has an initial object, namely $S = S[-0]^\sharp$. Therefore the ordinary limit of F over B_S is $F(S)$, and evaluation at an initial object is exact. Hence

$$R\varprojlim_{B_S} F \simeq F(S)$$

concentrated in degree 0. Thus every hypercover in the cofinal system computes the same complex $F(S)$, and the transition maps are the identity under this identification. Consequently

$$R\Gamma(A_S^{\mathrm{op}}, \pi_S^{-1}F) \simeq F(S).$$

The identification is the one induced by the adjunction unit, so the map $F(S) \to R\Gamma(A_S^{\mathrm{op}}, \pi_S^{-1}F)$ is a quasi-isomorphism. □

5 The global statement

Theorem 5.1. *Let P be a monoid. For every presheaf of abelian groups F on B_P^{op}, the adjunction morphism*

$$F \longrightarrow R\pi_*\pi^{-1}F$$

is a quasi-isomorphism.

Proof. Because the topology on B_P^{op} is trivial, quasi-isomorphisms of complexes of sheaves on B_P^{op} are detected objectwise.

Fix $b = P[-x]^\sharp \in B_P$. By Lemma 2.1,

$$(R\pi_*\pi^{-1}F)(b) \simeq R\Gamma\big(A_b^{\mathrm{op}}, \pi_b^{-1}(F|_{B_b})\big).$$

Applying Proposition 2.2 to the monoid b, we obtain

$$R\Gamma\big(A_b^{\mathrm{op}}, \pi_b^{-1}(F|_{B_b})\big) \simeq F(b)$$

concentrated in degree 0. Under this identification, the adjunction unit

$$F(b) \longrightarrow (R\pi_*\pi^{-1}F)(b)$$

is the identity. Since this holds for every $b \in B_P$, the morphism

$$F \longrightarrow R\pi_*\pi^{-1}F$$

is an objectwise quasi-isomorphism, hence a quasi-isomorphism of sheaves on B_P^{op}. □

References

[SGA4] M. Artin, A. Grothendieck, and J.-L. Verdier, *Théorie des topos et cohomologie étale des schémas*, Lecture Notes in Mathematics, Vols. 269, 270, 305, Springer-Verlag, 1972–1973.

[Qui73] D. Quillen, *Higher algebraic K-theory I*, in *Algebraic K-theory I: Higher K-theories*, Lecture Notes in Mathematics, Vol. 341, Springer-Verlag, 1973.

[KKMSD73] G. Kempf, F. Knudsen, D. Mumford, and B. Saint-Donat, *Toroidal Embeddings I*, Lecture Notes in Mathematics, Vol. 339, Springer-Verlag, 1973.

[Ogu18] A. Ogus, *Lectures on Logarithmic Algebraic Geometry*, Cambridge Studies in Advanced Mathematics, Vol. 178, Cambridge University Press, 2018.